

Effects of Novel Segmentation Framework ConvUNext Network along with Spatial Attention Module on Small Brain Tumor Dataset

Rambarki Pavan Kumar
Department of ECE
NIT Rourkela
Odisha, India, 769008
pavanrambariki@gmail.com

Pranshu Jena
Department of ECE
NIT Rourkela
Odisha, India, 769008
pranshujena16@gmail

Umesh C. Pati
Department of ECE
NIT Rourkela
Odisha, India, 769008
ucpati@nitrkl.ac.in

Abstract—The segmentation of brain tumor images has been essential for diagnosing the tumorous region. This helps in the development of effective treatment strategies and guiding surgical decisions. Manual segmentation methods had been used earlier, which led medical practitioners, researchers, and radiologists to recognize the tumors at a very later stage. The proposed segmentation framework has been trained, validated, and tested in a publicly available dataset that provides various MRI scans of glioma tumors at different stages. The proposed framework for brain tumor segmentation incorporates pre-trained ConvNeXt blocks as the backbone of the U-Net architecture, further enhanced by spatial attention modules (SAM). The proposed framework utilizes the pre-trained convolutional network ConvNeXt as the encoder to extract spatial features while efficiently addressing the vanishing gradient issues. The U-Net-based decoder is employed for precise semantic segmentation, ensuring high spatial accuracy, essential for segmenting the tumorous region. Spatial attention mechanisms are integrated across the encoder-decoder structure to focus on tumor-specific regions and suppress irrelevant areas, thereby improving the precision and overall efficiency of the proposed framework. BraTs 2020 brain T1-weighted MRI dataset has been used to perform segmentation in the proposed framework. The framework demonstrated outstanding performance with a Dice Score Coefficient(DSC) of 93.49% for ConvNeXt U-Net. The combination of the advanced feature extraction capabilities of ConvNeXt with attention-guided segmentation makes this framework outperform state-of-the-art models, offering superior segmentation accuracy. The findings highlight the potential of this approach in enhancing brain tumor segmentation for better disease understanding, diagnosis, and treatment planning.

Index Terms—Pre-trained Backbone, ConvNeXt, U-Net, Spatial attention module, Segmentation, Small dataset

I. INTRODUCTION

Brain tumors are an aggressive and abnormal expansion of brain tissue that has been a serious risk to human health through the ages[1]. An estimated 308,000 people are diagnosed with brain tumors worldwide every year, and gliomas constitute the largest proportion of primary malignant brain tumors[2]. These tumors are amongst the most aggressive and life-threatening of all tumors, resulting

in high mortality of the patient. The golden standard for brain tumor diagnosis is Magnetic Resonance Imaging (MRI) because it is a non-invasive imaging modality that provides excellent structural information at different sequences (T1, T2, FLAIR)[3]. MRI offers an effective method that aids in a thorough examination of the tumorous region in the brain. This requires an intervention of radiologists to identify brain tumors by their physical appearance[4]. Yet this method is very labor-intensive, time-consuming, and prone to error, especially in cases where tumors have irregular or diffuse borders. This fuels the demand for automated and semi-automated Computer-Aided Diagnosis (CAD) systems to improve diagnostic accuracy and efficiency[5].

To tackle labor-intensive, error-prone diagnosis problems, researchers have started using Machine Learning(ML) and, more recently, Deep Learning(DL) methods for separating brain tumors in MRIs [6]. Convolutional neural networks have become a top choice because they can automatically recognize important details from imaging data based on labels. Unlike older techniques, convolutional neural networks (CNNs) allow for direct learning from the data to minimize the time consumed on preprocessing. These improvements play an essential role in the detection framework that uses CNNs and have shown remarkable progress in segmenting tumors, leading to better accuracy, strength, and efficiency in analyzing brain tumors.

II. RELATED WORK

Akbar et al., [2] came up with a novel Single Level UNet3D design featuring a Multipath Residual Attention Block (MRAB) aimed at segmenting brain tumors. This approach makes use of atrous convolution and attention techniques in both the skip connections as well as processing blocks to boost how features are pulled out. The proposed model struggled with unseen data, making it difficult to generalize. Raza et al., [4] developed a 3D Deep Residual

U-Net (dResU-Net) model that processes multimodal MRI images for brain tumor segmentation. The 3D model paired a deep residual network as the encoder with the U-Net layout to address the vanishing gradients problem and kept the low-level features intact. In [3], the author proposed a method for segmenting brain tumors using the MedNeXt design, which draws inspiration from ConvNeXt. The novel MedNeXt architecture includes residual connections, depth-wise convolution with different large kernel sizes, and point-wise convolution with wider channels. The features generated had been classified with various machine learning classifiers. Khan et al., [5] proposed a novel state-of-the-art model, i.e., the Hybrid Attention-Based Residual U-Net (HA-RUNet) model, for segmenting brain tumors. The proposed segmentation model mixes lightweight modules for feature extraction, residual blocks, and attention methods to improve segmentation results while keeping computational demands reasonable. Liu et al., [6] proposed a novel architecture, SGEResU-Net, which is a 3D U-Net model that includes residual blocks and Spatial Group-wise Enhance (SGE) attention modules. This proposed architecture had been specifically designed for brain tumor segmentation and enhances feature learning while reducing noise. The proposed model achieved Dice Similarity Coefficients(DSC) and Hausdorff distances for segmenting the tumor and its sub-regions.

A. Novel Contribution

The main contributions of the proposed frameworks are as follows:

- 1) A robust U-Net-based framework contains backbone pre-trained CNN blocks and is enhanced with Spatial Attention Modules(SAM) to improve segmentation of brain tumors in T1-weighted MRI images, effectively capturing fine details and complex tumor boundaries.
- 2) Integration of pre-trained CNN blocks within the contracting pathway to address the vanishing gradient issue, enabling the network to extract both low-level and high-level features efficiently, ensuring better feature propagation and model stability.
- 3) Incorporation of spatial attention modules across the encoder-decoder structure to refine feature maps, focusing on critical regions during training by emphasizing spatially important areas, leading to more precise segmentation outputs.

The rest of the paper contains the proposed architecture in Section IV. The methodology in Section III contains the pre-processing of the dataset and proposed framework. The results obtained from the proposed model and discussion are contained in Section V. The conclusion and future scope have been provided in Section VI

III. METHODOLOGY

A. Dataset

The Brain Tumor Segmentation BraTS 2020 dataset [7] has been designed explicitly for brain tumor segmentation. The BraTS 2020 training dataset comprises 369 MRI scans across

TABLE I: Details of Brats2020 Dataset.

Dataset	Brats2020
Data format	3D NiftI format (. nii.gz files)
Image modalities	T1, T2, T1ce and Flair
Training data of T1 weighted	100 patients
No. of T1 slices per patient	155 slices
No. of mask slices per patient	155 slices
Image size	240×240

various modalities. Each dataset includes five Neuroimaging Informatics Technology Initiative (NiftI) files. These files provide the ground truth for the tumor and four MRI modalities, i.e., T1-weighted, T1-weighted contrast-enhanced (T1CE), T2-weighted, and T2-weighted fluid-attenuated inversion recovery (FLAIR) as shown in Table I. This work has considered only 100 T1-weighted NiftI patient scans of each of the 155 slices. However, in this study, the proposed framework focuses exclusively on a small dataset of T1-weighted images. The size of each slice is 240×240 for single-modality segmentation, as T1 images provide essential anatomical information. The slices of the brain tumor have been shown in Fig 1

B. Pre-processing

This study utilized a small portion of the BRaTS 2020 dataset of T1-weighted brain MRI scans comprising 100 patients. Each 3D MRI scan has been sliced into 155 2D scans, yielding 15,500 slices. The 2D slices were normalized to ensure consistent intensity distribution across all samples. The small brain tumor dataset has been utilized to optimize and fine-tune model performance efficiently, as smaller datasets reduce computational overhead during training. MRI images can vary significantly in intensity across different scans due to variations in MRI machines and protocols. This difference in intensities makes the DL model hard to compute without normalized value and uses high computational resources. To minimize the usage of computational resources and maintain consistency throughout the dataset, the MRI scans have been intensity-normalized by scaling pixel values to a standard range (e.g., between 0 and 1). The small dataset has been divided into 3 sets, i.e., training, Validation, and testing, into the ratio of 80:15:5, respectively.

C. Model Architecture

ConvNeXt is a pre-trained CNN that follows the principle of Vision Transformers' performance that maintains the cost-effectiveness of traditional CNN. Adding depthwise convolutions, replacing BatchNorm with LayerNorm, and layering features improves earlier CNN models. These enhancements make it better at image classification, segmentation, and object detection. This architecture is available in popular deep-learning programs like PyTorch and 'timm', providing a versatile and effective solution for current projects. The structure of the ConvNeXt backbone is shown in Fig.2.

The SAM improves segmentation by zeroing in on the important areas in feature maps. Spatial information is

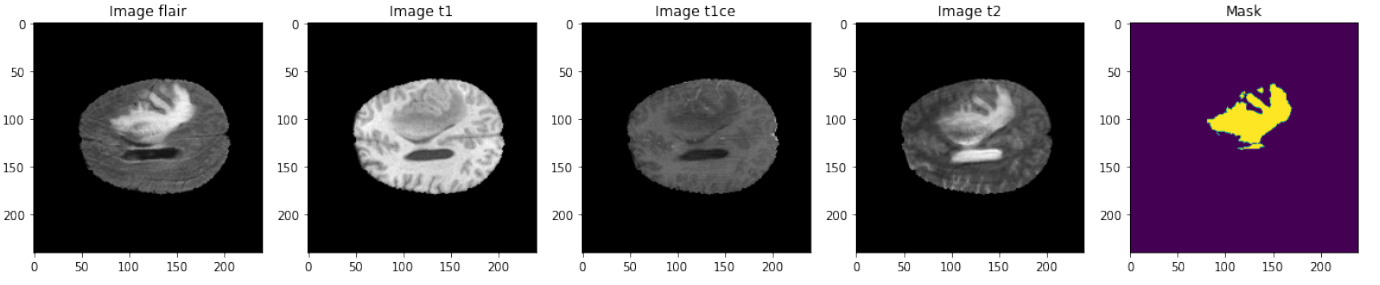


Fig. 1: Slices from Brats2020 with Different Modalities and its Mask

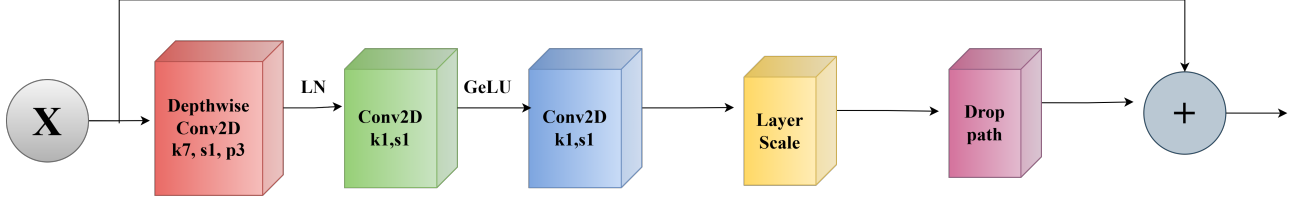


Fig. 2: ConvNeXt Backbone Structure

collected using average as well as max pooling and combined using a convolutional layer. This process creates an attention map, which is adjusted with a sigmoid function. SAM uses element-wise multiplication. The attention map highlights important sections and downplays less important ones. The module is lightweight and fits easily into models like U-Net and ConvNeXt, improving localization and segmentation with low computational expense.

U-Net, an encoder-decoder CNN, is used for semantic segmentation in biomedical image analysis. The encoder gradually decreases the input dimension to extract high-level information while the decoder upsamples to restore full resolution. Skip connections between the encoder and decoder preserve spatial features, improving segmentation accuracy. This architecture makes U-Net good in segmenting complex structures, especially in medical imaging with limited data. SAM helps U-Net to emphasize crucial places like brain tumors by focusing on the most important areas.

D. Focal Loss

The effectiveness and performance of deep learning models heavily depend on both the model architecture and the choice of loss function. Brain tumor segmentation poses significant challenges due to class imbalance, where some tumor regions occupy a much smaller area than other regions. To overcome this class imbalance, the focal loss function is introduced. Focal Loss is an improved variant of cross-entropy loss, created to address class imbalance by diminishing the impact of easily classified examples and concentrating on more challenging

ones. The formula for Focal Loss is given in *Equation(1)* :

$$F = \alpha \cdot (1 - p_t)^\gamma \cdot -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where: α is a balancing factor for class imbalance, p_t is the predicted probability for the true class, γ is the focusing parameter that down-weights easy examples, BCE Loss is the binary cross-entropy loss.

E. Evaluation Metrics

Evaluation of the segmentation performance of the proposed approach involves Dice Similarity Coefficient(DSC), Intersection of Union(IoU), accuracy, and precision. To evaluate the efficiency of the framework in the case of segmentation, the proportion of pixels that have been correctly classified for the area of interest according to the ground truth mask is calculated, which provides a measure of the model's accuracy in delineating the intended regions within the input data.

$$DSC = 2 \cdot \frac{|A_{pred} \cap A_{gt}|}{|A_{pred}| + |A_{gt}|} \quad (2)$$

In *Equation(2)* A_{pred} has been denoted as an area of the predicted region and A_{gt} has been denoted as the actual ground truth mask.

IoU measures the overlap divided by the total area of both the predicted mask and the ground truth mask of the tumorous regions.

$$IoU_i = \frac{TP_i}{TP_i + FP_i + FN_i} \quad (3)$$

In *Equation(3)*, it calculates the average of all the IoUs for $i = 1$ to the number of pixels in the object(the tumor region

of the brain).

Accuracy is expressed as in *Equation(4)*.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (4)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5)$$

Precision is expressed as in *Equation(5)*

IV. PROPOSED ARCHITECTURE

The proposed architecture for brain tumor segmentation enhances a U-Net framework with a SAM along with pre-trained ConvneXt as the backbone. The ConvneXt backbone uses the pre-trained weights and biases that enhance the performance of the proposed segmentation framework. This proposed segmentation framework focuses on critical spatial regions in the input feature maps, improving segmentation accuracy. The model processes single-channel T1 images from the BraTS2020 dataset, with the encoder extracting hierarchical features and SAM dynamically refining spatial information. Skip connections integrate low-level spatial details into the decoder, which reconstructs the segmentation mask through upsampling and feature fusion. A final 1×1 convolution produces the binary segmentation mask, with focal loss employed during training to address class imbalance and emphasize hard-to-segment regions. This design ensures precise delineation of tumor structures. The architecture is further enhanced through data augmentation and trained using optimizers like Adam or RMSprop to handle class imbalance and improve boundary delineation. This design effectively captures both global and local features, enabling accurate segmentation of complex tumor structures.

V. RESULTS AND ANALYSIS

A. Implementation Details

The programming language utilized was Python, and the model was developed using the PyTorch library. The RMSprop and Adam optimizers have been employed to train the model whereas Adam, is a quick-converging optimizer that adjusts learning rates through momentum and RMS, making it ideal for various tasks as well as RMSprop enhances training stability by normalizing learning rates according to recent gradient sizes, frequently favored for smaller datasets, with a learning rate of 0.001. Given the computational limitations, the model underwent training for 45 epochs with a batch size 16. The BraTS 2020 dataset has been utilized, following an 80:15:5 distribution for training, validation, and testing. The training is performed in an environment equipped with a GPU to optimize computational efficiency. Table II shows the software specifications.

TABLE II: Software specifications.

Software specifications	
Platform	Google Colab
Deep Learning Frame work	PyTorch
Programming language	Python 3.10.12
GPU	T4 GPU
CPU	Intel(R) Xeon(R) CPU @ 2.20GHz
RAM	12.6GB

B. Analysis of results

1) *Discussion:* The proposed segmentation framework model scored a Dice score of 0.9142 and demonstrated an IoU of 0.8420, which indicates a powerful feature extraction ability but is still limited. With the introduction of SAM, the performance significantly improved, and the proposed framework reached a Dice of 0.9349 and an IoU of 0.8777. The finding underscores the power of SAM in refining segmentation accuracy as it can prioritize spatially significant tumorous regions and capture fine-cut tumor details. The SAM also improved precision and overall accuracy. The model achieved a precision of 0.8848 and an accuracy of 0.9932 without SAM. Using SEM with the aforementioned metrics of 0.9351 and 0.9987 for valid metrics of SAM, It can be observed that SAM has an advantage in improving feature representation and reducing tumor segmentation false positives.

2) *Results:* The results obtained have been shown in Table III. The performance of the novel segmentation framework has been evaluated with and without the Spatial Attention Module (SAM), focusing on key metrics such as the Dice coefficient and Intersection over Union (IoU). The enhanced results that have been achieved by incorporating SAM show that attention mechanisms in brain tumor segmentation tasks lead to substantial improvement in the performance of the segmentation of the tumorous region. The comparison has been performed on a smaller section of the MRI dataset to finetune and make the model robust for unseen data. The semantic segmentation architecture, along with the ConvneXt backbone, efficiently captures the spatial domain features of the tumorous region with respect to the provided ground truth mask.

The improvement in segmentation results due to the use of SAM shows that the class imbalance caused by a large number of pixels of the non-tumorous region in pixel-wise classification has been addressed. This makes the model robust as well as provides a better performance in segemnting the tumorous region. The segmented output by the proposed model in different brain slices has been shown in Fig 5.

VI. CONCLUSION

This proposed framework provides an in-depth insight into the effects and benefits of the SAM on a small set of the BraTS dataset. The change in the backbone of the dataset massively improves the performance of the semantic segmentation. The CNN model provides pre-trained weights and biases that give good initialization to the CNN blocks of the UNet for

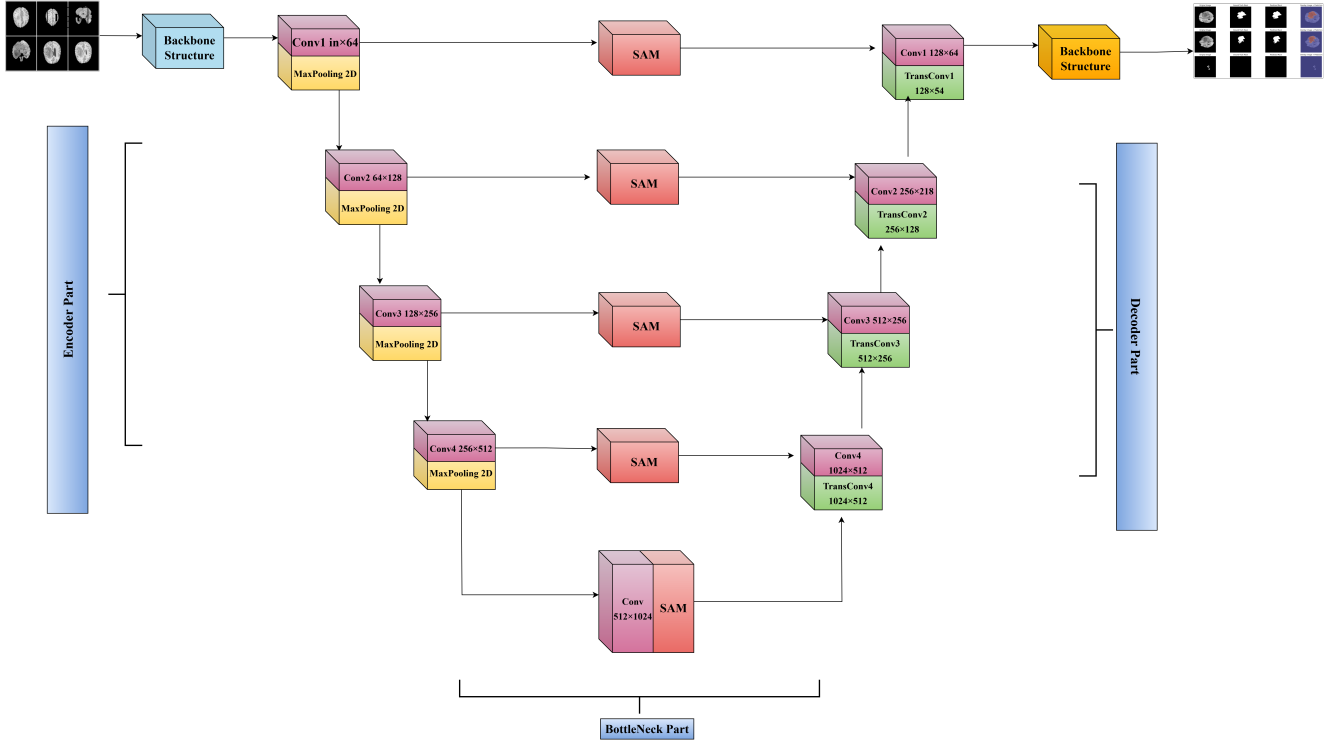


Fig. 3: Proposed Architecture

TABLE III: Evaluation of the Proposed Segmentation Framework.

Model	Loss Function	Optimizer	Loss	Dice Coefficient(%)	IoU (%)	Accuracy(%)	Precision(%)
ConvNext U-Net	Focal	Adam	1×10^{-5}	91.42	84.20	99.32	88.48
		RMS Prop	8×10^{-4}	88.61	78.53	99.08	89.24
ConvNeXt U-Net + SAM	Focal	Adam	2×10^{-4}	93.49	87.77	99.87	93.51
		RMS Prop	2×10^{-4}	91.42	84.19	99.55	91.21

TABLE IV: State-of-the-art(SOTA) of Different Segmentation Framework

Author	Method Used	DSC(%)
Liu et al., 2021. [6]	SGEResU-Net	90.10
Akbar et al.,2022.[2]	U-Net 3D with (MRAB)	89.59
Raza ei al.,2023. [4]	3D Residual U-Net	86.01
Khan et al., 2023. [5]	HA-ResU-Net	86.7
Maani et al., 2023.[3]	MedNext	85.54
proposed framework, 2025.	COnvNext U-Net + SAM	93.49

improved segmentation of the tumorous region. The novel framework used SAM to address class imbalance in pixel-wise classification. This framework aids in the automatic diagnosis of the tumorous region at different brain tumor stages. This framework not only provides a better result as compared to the state-of-the-art model but also addresses the class imbalance issue. The primary goal of this proposed segmentation framework is to perform better, which has been achieved with a DSC of 93.49%. In the future, the architecture can be improved and modified using different CNN models for better performance of segmenting the benign tumor and

differentiating it from the non-tumor region.

REFERENCES

- [1] P. Jena and U. C. Pati, "Performance study of optimizers for segmentation of brain tumors using atrous convolution in u-net," in *2023 International Conference on Communication, Circuits, and Systems (IC3S)*, 2023, pp. 1–5.
- [2] A. S. Akbar, C. Fatichah, and N. Suciati, "Single level unet3d with multipath residual attention block for brain tumor segmentation," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 6, Part B, pp. 3247–3258, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1319157822001069>
- [3] F. A. Maani, A. Ur Rehman Hashmi, N. Saeed, and M. Yaqub, "On enhancing brain tumor segmentation across diverse populations with convolutional neural networks," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024, pp. 1–4.
- [4] R. Raza, U. Ijaz Bajwa, Y. Mehmood, M. Waqas Anwar, and M. Hassan Jamal, "dresu-net: 3d deep residual u-net based brain tumor segmentation from multimodal

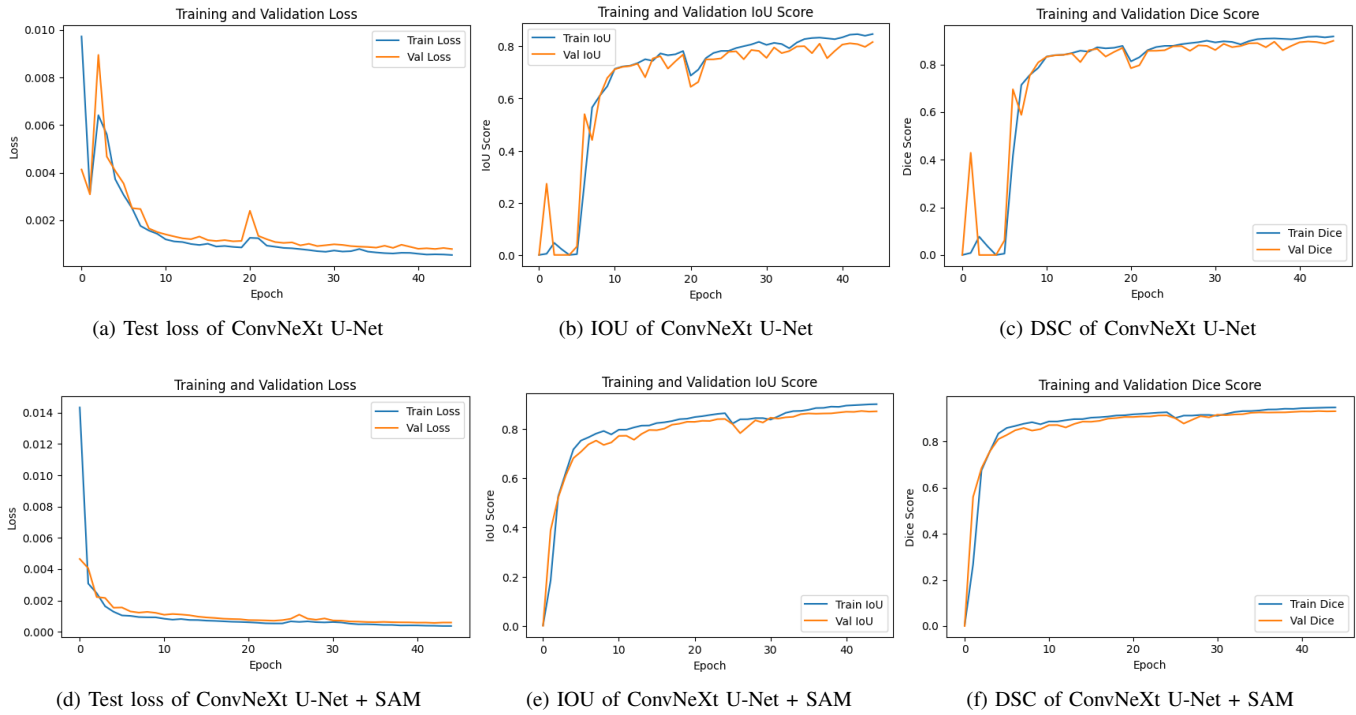


Fig. 4: Test loss, DSC and IoU graphs of training and validation results

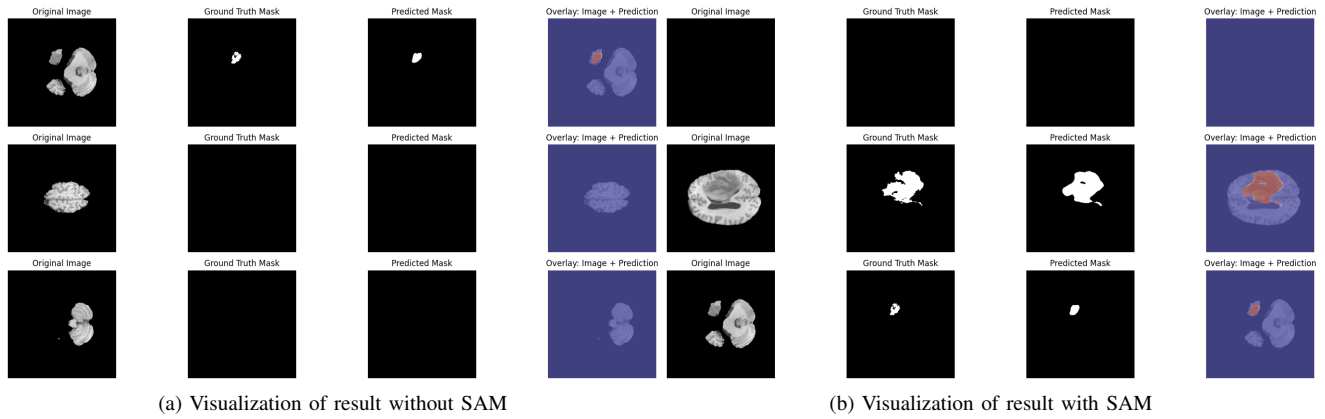


Fig. 5: Segmented region of the slices

- mri,” *Biomedical Signal Processing and Control*, vol. 79, p. 103861, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1746809422003809>
- [5] U. I. J. U. J. M. A. K. M. A. U. T. J.-H. C. Wajiha Rahim Khan, Tahir Mustafa Madni, “A hybrid attention-based residual unet for semantic segmentation of brain tumor,” *Computers, Materials & Continua*, vol. 76, no. 1, pp. 647–664, 2023. [Online]. Available: <http://www.techscience.com/cmc/v76n1/53090>
- [6] D. Liu, N. Sheng, T. He, W. Wang, J. Zhang, and J. Zhang, “Sgeresu-net for brain tumor segmentation,” *Mathematical Biosciences and Engineering*, vol. 19, no. 6, pp. 5576–5590, 2022. [Online]. Available: <https://www.aimspress.com/article/doi/10.3934/mbe.2022261>
- [7] B. H. Menze, A. Jakab, S. Bauer, J. Kalpathy-Cramer, K. Farahani, J. Kirby, Y. Burren, N. Porz, J. Slotboom, R. Wiest, L. Lanczi, E. Gerstner, M.-A. Weber, T. Arbel, B. B. Avants, N. Ayache, and t. y. n. p. k. d. Buendia, journal=IEEE Transactions on Medical Imaging.